Maui – Coastal flooding ESF-6 playbook

Data as of 2026-10-27

County: Maui · **Hazard:** Coastal flooding

Planning aid. Supports, does not replace, county emergency plans. Not an official shelter list.

1. Summary

People at risk **36,678**	Shelter demand (central) **1,834**	Shelter demand low – high **734 – 4,035**
Limited English (C) **1,041**	Power-dependent (M) **237**	Disability (I) **4,027**
Age 65+ (I) **7,035**	People without a vehicle (T) **1,598**	Known 2026 sites usable **6**

Shelter-seeking rates: low 2%, central 5%, high 11.0%. CMIST counts are people at risk, from census-tract estimates; one person can appear in more than one count.

2. Scope: county total vs. one community

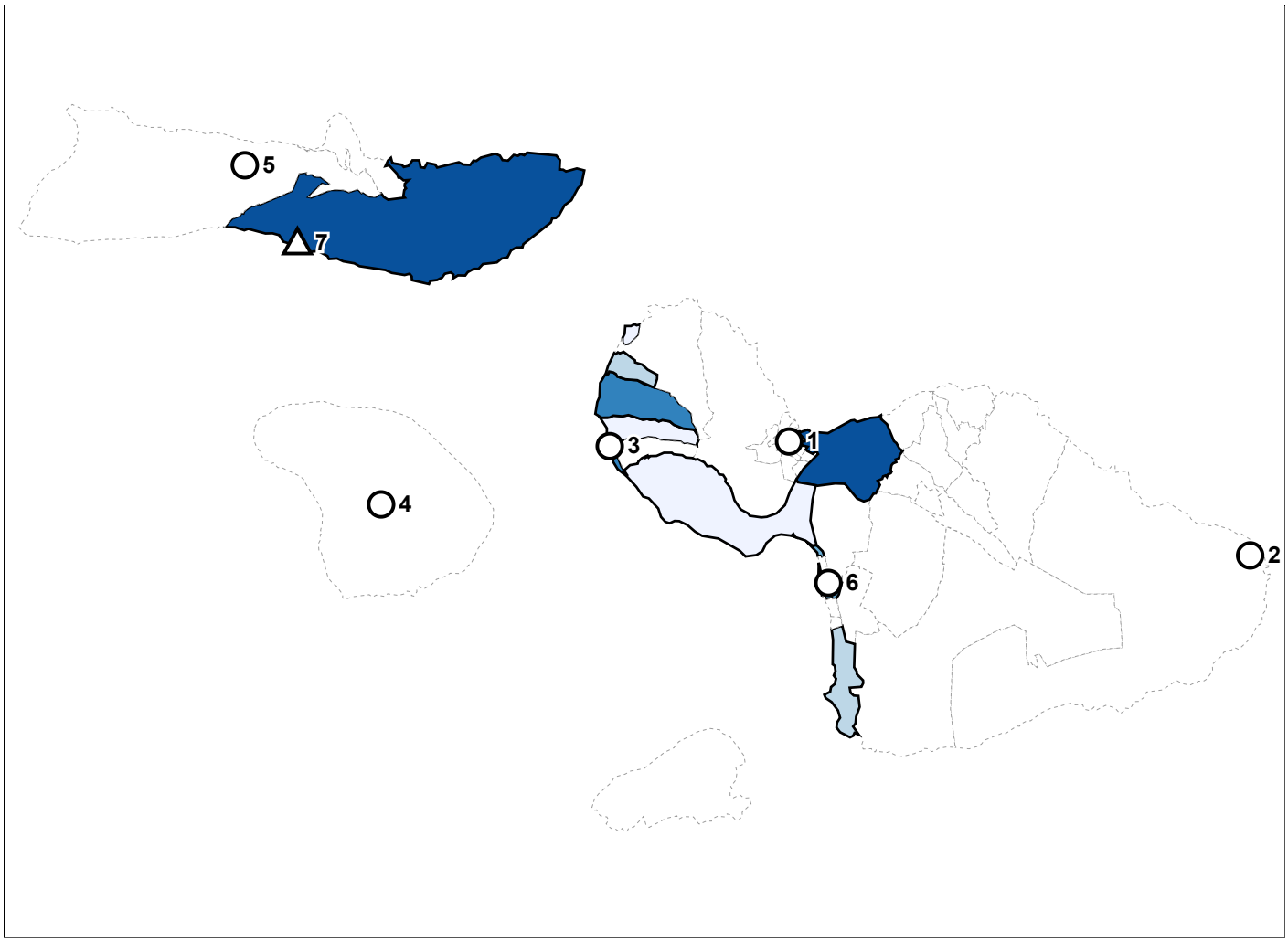
County total = upper bound. It assumes every at-risk tract is hit at once (top 25% of each county's tracts by NRI score). Real events usually hit fewer communities.

Largest single community (highest-demand census tracts)

Tract	At risk	Shelter low / central / high	Limited English	Power-dep.	Disability	65+	No vehicle
Tract 319 · Kahului	5,325	106 / **266** / 586	489	29	609	866	517
Tract 317 · near Kaunakakai	4,827	97 / **241** / 531	19	35	820	1,047	237
Tract 307.13 · Kihei	3,834	77 / **192** / 422	33	13	286	389	118

Plan a realistic one-community event with the first row.

3. Map: AFN need and known sites



- AFN need 0–318 people (class 1 of 5)
- AFN need 318–369 people (class 2 of 5)
- AFN need 369–450 people (class 3 of 5)
- AFN need 450–952 people (class 4 of 5)
- AFN need 952–1,644 people (class 5 of 5)
- Solid outline: tract in hazard footprint (at risk)
- Dashed outline: not in footprint
- Usable
- Check: flood (SFHA)
- Check: flood (tract-level)
- Flood risk undetermined (zone D)
- Excluded: tsunami zone

AFN need = power-dependent + disability + limited English + no-vehicle people at risk (people may be counted more than once). Numbers on markers match the sites table. No basemap: tract outlines only.

4. Known sites (opened in 2026 events)

#	Status	Site	Address	Reason	Times opened 2026
1	✓ **Usable**	Baldwin High School	1650 Kaahumanu Ave., Wailuku	FEMA zone X: outside Special Flood Hazard Area	1
2	✓ **Usable**	Hāna High & Elementary School Cafeteria	4111 Hāna Highway, Hāna	FEMA zone X: outside Special Flood Hazard Area	1
3	✓ **Usable**	Lahaina Civic Center Gym	1840 Honoapiʻilani Highway, Lahaina	FEMA zone X: outside Special Flood Hazard Area	2
4	✓ **Usable**	Lānaʻi Youth Center	101-165 Fifth St., Lānaʻi City	FEMA zone X: outside Special Flood Hazard Area	1
5	✓ **Usable**	Molokaʻi Baptist Church	495 Puupeelua Ave., Hoolehua	FEMA zone X: outside Special Flood Hazard Area	1
6	✓ **Usable**	South Maui Community Park Gym	1501 Liloa Drive, Kīhei	FEMA zone X: outside Special Flood Hazard Area	2
7	⚠ **Check – flood**	Kilohana Recreation Center	344 Kamehameha V Highway, Kaunakakai	FEMA Special Flood Hazard Area (zone AE)	1
8	**Not located**	Mayor Hannibal Tavares Community Center (Social Hall)	91 Pukalani St., Pukalani	not located – not on map	1

Candidate public schools in this county usable for this hazard: **32**. Sites are places a county has opened before; not every site opens in every event. Always follow current county announcements.

5. Needs and proposed partners

Central estimate. Roles are **proposed – to be confirmed with partner**. Pledged capacity stays blank until a partner confirms it.

Need	Quantity	Unit	Proposed lead	Proposed support	Pledged	Gap
Cots / sleeping space	1,834	cots	county_em	dhs; hiema; doe; hspls; red_cross; faith_based; other_agencies	—	confirm with partner
Floor space (≤72 h)	36,678	sq ft	county_em	dhs; hiema; doe; hspls; red_cross; faith_based; other_agencies	—	confirm with partner
Floor space (≤2 weeks)	73,356	sq ft	county_em	dhs; hiema; doe; hspls; red_cross; faith_based; other_agencies	—	confirm with partner
Water	1,834	gal/day	red_cross	dhs; county_em; doe; salvation_army; hawaii_voad; faith_based; business_community	—	confirm with partner
Meals	3,668	meals/day	red_cross	dhs; county_em; doe; salvation_army; hawaii_voad; faith_based; business_community	—	confirm with partner
Toilets (≤72 h)	46	toilets	county_em	dhs; hiema; doe; hspls; red_cross; faith_based; other_agencies	—	confirm with partner
Toilets (≤2 weeks)	92	toilets	county_em	dhs; hiema; doe; hspls; red_cross; faith_based; other_agencies	—	confirm with partner
Shelter staff	126	staff per shift	county_em	dhs; hiema; doe; hspls; red_cross; faith_based; other_agencies; hawaii_voad; oha; salvation_army; philanthropy; business_community	—	confirm with partner
Power outlets for power-dependent residents	12	outlets	doh	county_em; health_care; utilities	—	confirm with partner
Accessible transport	1,598	people in no-vehicle households at risk	county_em	dhs; hiema; dot; business_community; other_agencies	—	confirm with partner
Pet space – dog households	285	households with a dog	county_em	hawaii_voad	—	confirm with partner
Pet space – cat households	166	households with a cat	county_em	hawaii_voad	—	confirm with partner
Children's area	361	children in shelter	county_em	dhs; hiema; doe; hspls; red_cross; faith_based; other_agencies; doh; health_care; philanthropy	—	confirm with partner
Interpreters – Ilocano	22	people with limited English in shelter	dhs	county_em; hiema; hspls; doh; oha; faith_based; other_agencies	—	confirm with partner
Interpreters – Tagalog, Filipino	12	people with limited English in shelter	dhs	county_em; hiema; hspls; doh; oha; faith_based; other_agencies	—	confirm with partner
Interpreters – Spanish	4	people with limited English in shelter	dhs	county_em; hiema; hspls; doh; oha; faith_based; other_agencies	—	confirm with partner
Interpreters – Japanese	3	people with limited English in shelter	dhs	county_em; hiema; hspls; doh; oha; faith_based; other_agencies	—	confirm with partner
Interpreters – Marshallese	1	people with limited English in shelter	dhs	county_em; hiema; hspls; doh; oha; faith_based; other_agencies	—	confirm with partner
Interpreters – Hawaiian	1	people with limited English in shelter	dhs	county_em; hiema; hspls; doh; oha; faith_based; other_agencies	—	confirm with partner

6. Decision points

Proposed – supports, does not replace, county plans.

When	Trigger	Action	Owner (proposed)	Support
BEFORE	Annual pre-positioning review (no official product) County (sets the date)	Mark coastal shelters in V/A zones for site check; choose inland alternates.	County Emergency Management (each county)	doe, dhs
BEFORE	Coastal Flood Warning NOAA NWS Honolulu	Alert priority coastal tracts; confirm inland shelters and transport.	County Emergency Management (each county)	dot, dhs
AFTER	Warning cancelled or all clear County / NWS / PTWC	Recovery sheltering and feeding for displaced households.	American Red Cross	hawaii_voad, salvation_army

7. Priority outreach and languages

Top 10 census tracts by AFN need (tract-level counts only)

#	Tract	At risk	Power-dep.	Disability	Limited English	No vehicle	AFN need
1	Tract 319 · Kahului	5,325	29	609	489	517	1,644
2	Tract 317 · near Kaunakakai	4,827	35	820	19	237	1,111
3	Tract 315.03 · Kaanapali	2,834	21	403	143	385	952
4	Tract 314.04 · Lahaina	3,100	23	306	97	123	550
5	Tract 307.13 · Kihei	3,834	13	286	33	118	450
6	Tract 307.08 · Kihei	2,881	17	345	82	0	444
7	Tract 307.06 · Kihei	2,796	13	350	25	11	399
8	Tract 315.05 · Mahinahina	2,341	15	178	31	145	370
9	Tract 303.07 · Makena / Wailea	1,903	30	285	28	14	357
10	Tract 314.02 · Lahaina	3,364	16	224	62	16	318

Languages: people who speak English less than "very well" (Kauai + Maui)

Kaua‘i and Maui share one language list: ACS PUMS reports both counties (and Kalawao) as a single area, PUMA 00100, so they cannot be separated.

Language	People 5+	90% MOE
Ilocano	9,131	±1,439
Tagalog, Filipino	5,074	±926
Spanish	1,711	±554
Japanese	1,339	±417
Marshallese	633	±488
Hawaiian	515	±289

Source: ACS PUMS 2019-2023 5-year (validated against DBEDT Data Book 2024, Table 1.44). Pre-approved message templates: none yet; no machine translation without human review.

8. Sources and assumptions

Planning factors

Factor	Value	Source or team assumption
Shelter-seeking rate – Hurricane	0.02 / 0.03 / 0.05	South Carolina EMD, 2026 Hurricane Response Annex G (planning factors 2%, 3%, 5%)
Shelter-seeking rate – Wildfire	0.02 / 0.05 / 0.112	high = observed Maui 2023: 1,418 people in shelters on Aug 11, 2023 (Maui Now) / Lahaina CDP population 12,702 (2020 Census) = 0.1116, rounded to 3 decimals = 0.112 (so the calibration event falls inside the band). low and central are team_assumption.
Shelter-seeking rate – Tsunami	0.02 / 0.05 / 0.11	team_assumption: central 5%; low 2% = hurricane planning factor (SCEMD); high 11% = observed Maui 2023 no-notice total-loss rate
Shelter-seeking rate – Inland flooding	0.02 / 0.05 / 0.11	team_assumption: central 5%; low 2% = hurricane planning factor (SCEMD); high 11% = observed Maui 2023 no-notice total-loss rate
Shelter-seeking rate – Coastal flooding	0.02 / 0.05 / 0.11	team_assumption: central 5%; low 2% = hurricane planning factor (SCEMD); high 11% = observed Maui 2023 no-notice total-loss rate
Footprint – Hurricane	all	Hurricanes affect whole islands (Lala, Lowell 2026).
Footprint – Tsunami	tsunami_zone_share	Population share inside the State standard tsunami evacuation zone. Extreme zone is reported separately.
Footprint – Wildfire	nri_top_share_by_county (25%)	Counties lead mass care, so each county plans for its own highest-risk tracts (team_assumption).
Footprint – Inland flooding	nri_top_share_by_county (25%)	Counties lead mass care, so each county plans for its own highest-risk tracts (team_assumption).
Footprint – Coastal flooding	nri_top_share_by_county (25%)	Counties lead mass care, so each county plans for its own highest-risk tracts (team_assumption).
floor sqft 72h	20	Florida DEM Sheltering Guidance Aid (2010): emergency evacuation shelter, <=72 h
floor sqft 2wk	40	Florida DEM (2010): short-term shelter, <=2 weeks
water gal per day	1	Florida DEM (2010)
meals per day	2	Florida DEM (2010): two hot meals a day plus snacks
persons per toilet short term	20	Florida DEM (2010): short-term shelter, <=2 weeks
power outlets per power dependent person	1	team_assumption
persons per toilet 72h	40	Florida DEM (2010): emergency evacuation shelter, <=72 h
Staffing ratios	client registration 1:165, dormitory management 1:100, meal service 1:110, mental health 1:250, public health medical 1:200, security 1:100, sanitation 1:200, children area 1:12	One staff member per N shelter residents, minimum 1 when the shelter is open. Florida DEM (2010) staffing metrics; the source marks them as recommendations.
dog owning household share	0.446	AVMA U.S. pet ownership statistics (2020 data). Verified in 2022 AVMA Pet Ownership and Demographics Sourcebook, p. 3: 'Dog ownership increased by 6% from 38.4% (2016) to 44.6% (2020)'. The first url did not resolve when checked 2026-10-27.
cat owning household share	0.26	AVMA (2020 data). Verified in 2022 AVMA Pet Ownership and Demographics Sourcebook, p. 3: 'Cat ownership increased slightly from 25% in 2016 to 26% in 2020'.

Data sources

Source	Fetched	URL / file
CDC/ATSDR Social Vulnerability Index 2022, census tracts	2026-10-27	https://onemap.cdc.gov/onemapservices/rest/services/SVI/CDC_ATSDR_Social_Vulnerability_Index_2022_USA/FeatureServer/2

Source	Fetched	URL / file
FEMA National Risk Index v1.20, census tracts – Hurricane	2026-10-27	https://services.arcgis.com/XG15cJAlne2vxtgt/arcgis/rest/services/National_Risk_Index_Census_Tract_Hurricane_Hazard_Type_Risk_Index_
FEMA National Risk Index v1.20, census tracts – Wildfire	2026-10-27	https://services.arcgis.com/XG15cJAlne2vxtgt/arcgis/rest/services/National_Risk_Index_Census_Tract_Wildfire_Hazard_Type_Risk_Index_Ra
FEMA National Risk Index v1.20, census tracts – Tsunami	2026-10-27	https://services.arcgis.com/XG15cJAlne2vxtgt/arcgis/rest/services/National_Risk_Index_Census_Tract_Tsunami_Hazard_Type_Risk_Index_R
FEMA National Risk Index v1.20, census tracts – Inland Flooding	2026-10-27	https://services.arcgis.com/XG15cJAlne2vxtgt/arcgis/rest/services/National_Risk_Index_Census_Tract_Inland_Flooding_Hazard_Type_Risk_I
FEMA National Risk Index v1.20, census tracts – Coastal Flooding	2026-10-27	https://services.arcgis.com/XG15cJAlne2vxtgt/arcgis/rest/services/National_Risk_Index_Census_Tract_Coastal_Flooding_Hazard_Type_Risk_
Tsunami Evacuation Zones (Statewide GIS)	2026-10-27	https://geodata.hawaii.gov/arcgis/rest/services/Hazards/MapServer/2
Extreme Tsunami Evacuation Zones (Statewide GIS)	2026-10-27	https://geodata.hawaii.gov/arcgis/rest/services/Hazards/MapServer/12
HHS emPOWER – electricity-dependent DME, county level	2026-10-27	https://services2.arcgis.com/ZQ4jTQn6k7VPXEwO/arcgis/rest/services/HHS_emPOWER_REST_Service_Public/FeatureServer/2
Hawaii Statewide GIS – State DFIRM (FEMA flood hazard areas, 6/15/26)	2026-10-27	https://geodata.hawaii.gov/arcgis/rest/services/Hazards/MapServer/6

Source	Fetched	URL / file
County shelter activations 2026 (official county releases)	see table	data/sources/county_shelter_activations.csv
Census TIGERweb 2020 Census Designated Places (tract community labels)	2026-10-27	https://tigerweb.geo.census.gov/arcgis/rest/services/TIGERweb/tigerWMS_Census2020/MapServer/28
ACS PUMS 2019-2023 5-year (detailed languages)	2026-10-27	https://www2.census.gov/programs-surveys/acs/data/pums/2023/5-Year/csv_phi.zip
DBEDT State of Hawaii Data Book 2024, Table 1.44 (validation)	2026-10-27	https://files.hawaii.gov/dbedt/economic/databook/2024-individual/01/014424.pdf
Partner list: Hawaii DHS, HAIC 2026 ESF-6 challenge statement	2026-10-15	HAIC_AI Challenge_Submission_DHS_Disaster_2026.pdf
pre season hurricane		https://www.nhc.noaa.gov/cphcmission.php
hurricane watch		https://www.nhc.noaa.gov/aboutgloss.shtml
hurricane warning		https://www.nhc.noaa.gov/aboutgloss.shtml
tsunami warning		https://www.tsunami.gov/?page=message_definitions
flash flood warning		https://www.weather.gov/safety/flood-watch-warning
red flag warning		https://www.weather.gov/safety/wildfire-ww
coastal flood warning		https://www.weather.gov/okx/marinedef

Privacy: aggregate census-tract and county data only. No individual records. Public facility names and addresses only.